

Data Mining

DATA ANALYSIS

BY: IVETA MRÁZOVÁ

DATA ANALYSIS

Contents:

- Data standardization techniques
- Range, quartiles, and outliers
- Relationship between the attributes
- Discriminant analysis
- Cluster analysis

Contents:

- Data standardization techniques
- Range, quartiles, and outliers
- Relationship between the attributes
 - Relationship between categorical variables
 - χ^2 - test and Fisher's test
 - Regression analysis (~ relationship between numerical variables)
 - Linear regression for one input variable, correlation coefficients
 - Multi-dimensional linear regression
 - Logistic regression
- Discriminant analysis
- Cluster analysis
 - The main clustering approaches
 - Distance measures
 - k-means clustering
 - Hierarchical clustering
 - Vector quantization
 - Other clustering techniques
 - k-medoids algorithm, *grid-based methods*, and *density-based algorithms*
 - Scalable data clustering approaches

DATA ANALYSIS

Contents:

- **Data standardization techniques**
- *Range, quartiles, and outliers*
- *Relationship between the attributes*
- *Discriminant analysis*
- *Cluster analysis*

Contents:

- **Data standardization techniques**
- *Range, quartiles, and outliers*
- *Relationship between the attributes*
 - *Relationship between categorical variables*
 - χ^2 - test and Fisher's test
 - *Regression analysis (~ relationship between numerical variables)*
 - *Linear regression for one input variable, correlation coefficients*
 - *Multi-dimensional linear regression*
 - *Logistic regression*
- *Discriminant analysis*
- *Cluster analysis*
 - *The main clustering approaches*
 - *Distance measures*
 - *k-means clustering*
 - *Hierarchical clustering*
 - *Vector quantization*
 - *Other clustering techniques*
 - *k-medoids algorithm, grid-based methods, and density-based algorithms*
 - *Scalable data clustering approaches*

Data Standardization

- In the Euclidean space, standardization of attributes is recommended so that all attributes can have equal impact on the computation of distances.
- Example: Consider the following pair of data points
 - $x_i: (0.1, 20)$ and $x_j: (0.9, 720)$,
$$\text{dist}(x_i, x_j) = \sqrt{(0.9 - 0.1)^2 + (720 - 20)^2} = 700.000457,$$
- The distance is dominated by $(720-20) = 700$.
- **Standardize attributes:** to force the attributes to have a common value range

Interval-Scaled Attributes

- Their values are real numbers following a linear scale.
 - e.g., the difference in Age between 10 and 20 is the same as that between 40 and 50.
 - The key idea is that intervals keep the same importance throughout the scale
- **Decimal scaling** to the interval $[-1,1]$: divides the attribute values by the smallest power of 10 that keeps all the transformed values within the interval $[-1,1]$.
- **Range standardization** to the interval $[0,1]$ normalizes attribute values in the following way (f is an attribute):

$$range(x_{if}) = \frac{x_{if} - \min(f)}{\max(f) - \min(f)},$$

Interval-Scaled Attributes (cont. ...)

- **Range standardization** to the interval $[-1,1]$ normalizes attribute values in the following way (using $range(x_{if})$ defined above):

$$range^{[-1,1]}(x_{if}) = 2 \cdot range(x_{if}) - 1$$

- **Standardization according to the mean absolute difference** transforms the attribute values so that they have a zero mean and a mean absolute difference equal to 1.
 - For attribute f , the mean absolute difference (denoted by s_f) is computed as:

$$m_f = \frac{1}{n} (x_{1f} + x_{2f} + \dots + x_{nf}),$$

$$s_f = \frac{1}{n} (|x_{1f} - m_f| + |x_{2f} - m_f| + \dots + |x_{nf} - m_f|),$$

Score: $z(x_{if}) = \frac{x_{if} - m_f}{s_f}.$

Interval-Scaled Attributes (cont. ...)

- **Standardization according to the standard deviation** transforms the attribute values so that they have zero mean and (corrected) sample standard deviation equal to 1.
 - The (corrected) sample standard deviation of attribute f (denoted by $^{cor}s_f$) is computed as follows:

$$m_f = \frac{1}{n} (x_{1f} + x_{2f} + \dots + x_{nf}),$$

$$^{cor}s_f = \sqrt{\frac{1}{n-1} \left((x_{1f} - m_f)^2 + (x_{2f} - m_f)^2 + \dots + (x_{nf} - m_f)^2 \right)},$$

$$\text{std-score: } \text{std}(x_{if}) = \frac{x_{if} - m_f}{^{cor}s_f}.$$

Ratio-scaled attributes

- Numeric attributes, but unlike interval-scaled attributes, their scales are exponential,
- For example, the total amount of microorganisms that evolve in time t is approximately given by Ae^{Bt} , where A and B are some positive constants.

=> first, perform a log transform: **$\log(x_{if})$** ,

then, treat the transformed attribute values like interval-scaled attribute values

Nominal attributes

- Sometimes, we need to transform nominal attributes (e.g., colors, kinds of fruits, etc.) to numeric attributes

=> Transform nominal attributes to binary attributes:

- Let the number of values of a nominal attribute be v .
- Create v binary-valued attributes to represent them.
- If a data instance for the nominal attribute takes a particular value, the value of its binary attribute is set to 1, otherwise it is set to 0.
- The resulting binary attributes can be used as numeric attributes, with two values, 0 and 1.

Nominal attributes: an example

- Nominal attribute *fruit* has three values:
 - *Apple*, *Orange*, and *Pear*
- ==> **We create three binary attributes** - called, ***Apple***, ***Orange***, and ***Pear*** in the new data.
- If a particular data instance in the original data has *Apple* as the value for *fruit*,
 - then in the transformed data, we **set the value of the attribute *Apple* to 1**, and
 - **the values of attributes *Orange* and *Pear* to 0**

Ordinal attributes

- Ordinal attribute: an ordinal attribute is like a nominal attribute, but its values have a numerical ordering.
 - Example: Age attribute with the values Young, MiddleAge and Old. These values are ordered.

=> Common approach to standardization:
treat ordinal attributes as interval-scaled attribute.

DATA ANALYSIS

Contents:

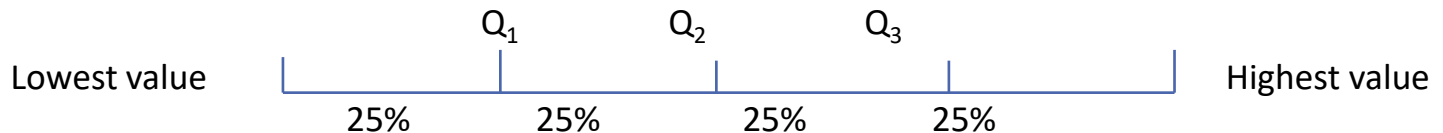
- *Data standardization techniques*
- **Range, quartiles, and outliers**
- *Relationship between the attributes*
- *Discriminant analysis*
- *Cluster analysis*

Contents:

- *Data standardization techniques*
- **Range, quartiles, and outliers**
- *Relationship between the attributes*
 - *Relationship between categorical variables*
 - χ^2 - test and Fisher's test
 - *Regression analysis (~ relationship between numerical variables)*
 - *Linear regression for one input variable and correlation coefficients*
 - *Multi-dimensional linear regression*
 - *Logistic regression*
- *Discriminant analysis*
- *Cluster analysis*
 - *The main clustering approaches*
 - *Distance measures*
 - *k-means clustering*
 - *Hierarchical clustering*
 - *Vector quantization*
 - *Other clustering techniques*
 - *k-medoids algorithm, grid-based methods, and density-based algorithms*
 - *Scalable data clustering approaches*

Range, Quartiles and Outliers

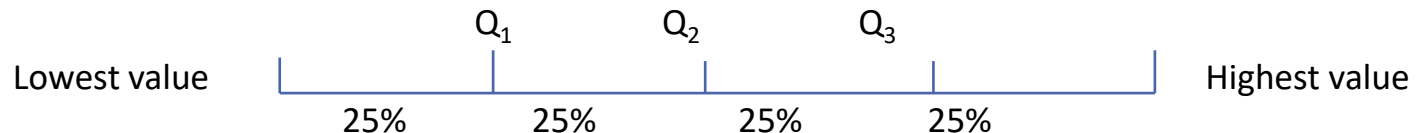
- The **range** of a set of data is the difference between the highest and the lowest value in the set.
- The quartiles Q_1 , Q_2 , Q_3 split the data into four parts



- 25% of the observations should have a value less than the lower quartile, Q_1
 - 50% of the observations should have a value below the median, Q_2 , and
 - 75% of the observations should have the value below the upper quartile, Q_3
- **Interquartile range** = upper quartile – lower quartile

Range, Quartiles and Outliers

How to calculate the quartiles for n ascending values:



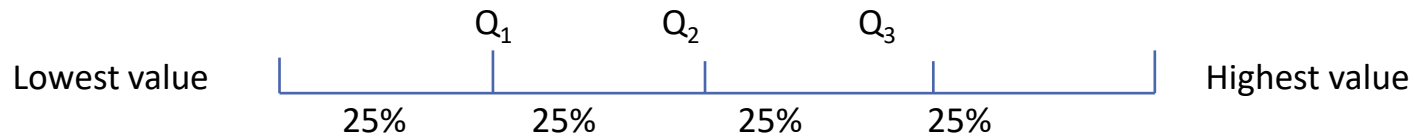
Discrete data:

- Lower quartile, Q_1 - divide n by **4**:
 - When $n/4$ is a whole number, find the midpoint of the corresponding term and the term above
 - When $n/4$ is not a whole number, round the number up and pick the corresponding term
- Upper quartile, Q_3 – divide n by **4** and multiply by **3**:
 - when $3n/4$ is a whole number, find the midpoint of the corresponding term and the term above
 - when $3n/4$ is not a whole number, round the number up and pick the corresponding term

Continuous data:

- Lower quartile, Q_1 :
 - Divide n by **4** and use interpolation to find the corresponding value
- Upper quartile, Q_3 :
 - Divide n by **4**, then multiply by **3** and use interpolation to find the corresponding value

Range, Quartiles and Outliers



- An **outlier** is an extreme value that lies outside the overall pattern of data.
- The following rule is often used to identify outliers:
An outlier is any value, which is
 - Greater than the upper quartile + $1.5 \cdot$ interquartile range or
 - Less than the lower quartile - $1.5 \cdot$ interquartile range

Range, Quartiles and Outliers

Example 1:

The blood glucose of 30 females is recorded. The results, in mmol/liter, are shown in **the stem and leaf diagram** below ($30/4 = 7.5$; 8th nr. is 3.2; 23rd nr. is 4.0).

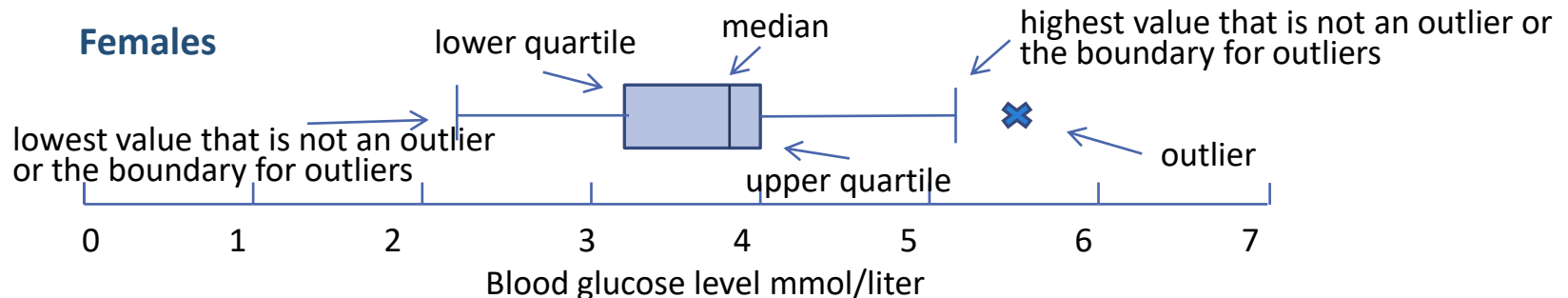
stem	leaf	Key: 2 1 means 2.1	
2	2 2 3 3 5 7		(6)
3	1 2 6 7 7 7 8 8 8 8 9 9 9		(13)
4	0 0 0 0 4 5 6 7 8		(9)
5	1 5		(2)

$$\begin{aligned} Q_1 &= 3.2 & Q_2 &= 3.8 \\ Q_3 &= 4.0 & \text{IQR} &= 0.8 \end{aligned}$$

$$\begin{aligned} L_{\text{limit}} &= 3.2 - 1.5 \cdot 0.8 = 2 \\ U_{\text{limit}} &= 4.0 + 1.5 \cdot 0.8 = 5.2 \end{aligned}$$

An outlier is the value 5.5.

Draw a box plot to represent the data. (The outlier is marked with a cross. The highest value which is not an outlier is 5.1. Always use a scale and label it. Give your box plot a title.)



Range, Quartiles and Outliers

Example 2:

Male and female turtles were weighted in grams. For males, the lower quartile was 400 g and the upper quartile was 580 g. For females, the lower quartile was 260 g and the upper quartile was 340 g.

An outlier is an observation that falls either 1x (interquartile range) above the upper quartile or 1 x (interquartile range) below the lower quartile.

- a) Which of these male turtles' weights would be outliers? (400 g, 260 g, 550 g, 640 g)

$$Q_1^M = 400 \mid Q_3^M = 580 \mid IQR^M = 180 \mid L_{\text{limit}} = 400 - 1 \cdot 180 = 220 \mid U_{\text{limit}} = 580 + 1 \cdot 180 = 760$$

Answer: There are no outliers among male turtles' weights.

- b) Which of these female turtles' weights would be outliers? (170 g, 300 g, 340 g, 440 g)

$$Q_1^F = 260 \mid Q_3^F = 340 \mid IQR^F = 80 \mid L_{\text{limit}} = 260 - 1 \cdot 80 = 180 \mid U_{\text{limit}} = 340 + 1 \cdot 80 = 420$$

Answer: There are the following outliers among female turtles' weights: 170 g and 440 g.

- c) What is the largest size a male turtle can be without being an outlier?

Answer: The largest size of a male turtle without being an outlier is 760 g.

DATA ANALYSIS

Contents:

- *Data standardization techniques*
- *Range, quartiles, and outliers*
- **Relationship between the attributes**
- *Discriminant analysis*
- *Cluster analysis*

Contents:

- *Data standardization techniques*
- *Range, quartiles, and outliers*
- **Relationship between the attributes**
 - **Relationship between categorical variables**
 - **χ^2 - test and Fisher's test**
 - *Regression analysis (~ relationship between numerical variables)*
 - *Linear regression for one input variable and correlation coefficients*
 - *Multi-dimensional linear regression*
 - *Logistic regression*
- *Discriminant analysis*
- *Cluster analysis*
 - *The main clustering approaches*
 - *Distance measures*
 - *k-means clustering*
 - *Hierarchical clustering*
 - *Vector quantization*
 - *Other clustering techniques*
 - *k-medoids algorithm, grid-based methods, and density-based algorithms*
 - *Scalable data clustering approaches*

Contingency Tables

~ relationship between two categorical quantities,
e.g., binary ones

A general contingency table

For n observations with
 R values for the quantity
 X and S values for the
quantity Y

	Y_1	Y_2	...	Y_S	Σ
X_1	a_{11}	a_{12}	...	a_{1S}	r_1
X_2	a_{21}	a_{22}	...	a_{2S}	r_2
...	...	...		...	...
X_R	a_{R1}	a_{R2}	...	a_{RS}	r_R
Σ	s_1	s_2	...	s_S	n

Contingency Tables (2)

	Y_1	Y_2	...	Y_S	Σ
X_1	a_{11}	a_{12}	...	a_{1S}	r_1
X_2	a_{21}	a_{22}	...	a_{2S}	r_2
...	...	...		...	...
X_R	a_{R1}	a_{R2}	...	a_{RS}	r_R
Σ	s_1	s_2	...	s_S	n

a_{kl} ... frequency of combination
 $(X = X_k) \wedge (Y = Y_l)$

r_k, s_l ... row / column totals
 (the so-called marginal values)

e_{kl} ... expected frequency of
 combination $(X = X_k) \wedge (Y = Y_l)$
 when assuming independence
 of X and Y

$$r_k = \sum_{l=1}^S a_{kl} \quad ; \quad s_l = \sum_{k=1}^R a_{kl} \quad ; \quad n = \sum_{k=1}^R \sum_{l=1}^S a_{kl} \quad ; \quad e_{kl} = \frac{r_k \cdot s_l}{n}$$

χ^2 – test

Checking the relationship between **X** and **Y**

Evaluation of the difference between the observed frequencies of the respective combinations (denoted in the table) and the frequencies expected if the hypothesis about the independence of both quantities (computed from the marginal values) would hold true

$$\chi^2 = \sum_{k=1}^R \sum_{l=1}^S \frac{(a_{kl} - e_{kl})^2}{e_{kl}} \Rightarrow \chi^2 = n \sum_{k=1}^R \sum_{l=1}^S \frac{\left(a_{kl} - \frac{r_k \cdot s_l}{n}\right)^2}{r_k \cdot s_l}$$

alternatively: $\chi^2 = \sum_{k=1}^R \sum_{l=1}^S \frac{a_{kl}^2}{e_{kl}} - n$

χ^2 – test (2)

- With the validity of the null hypothesis of the independence of the quantities X and Y :

$$H_0: P(X = X_k \wedge Y = Y_l) = P(X = X_k) P(Y = Y_l); \forall k, l$$

χ^2 distribution has $(R - 1) \cdot (S - 1)$ degrees of freedom

- If the value of the χ^2 –statistic is $\geq$ to the value of the χ^2 – distribution with the corresponding number of degrees of freedom at the chosen level of significance α :

$$\chi^2 \geq \chi^2_{(R-1)(S-1)}$$

the null hypothesis will be rejected

= > an alternative hypothesis of dependence is then accepted

χ^2 – test (3)

Example: a four-field contingency table

		PROVISION OF CREDIT		
		YES	NO	Σ
INCOME HEIGHT	High	50	0	50
	Low	30	40	70
	Σ	80	40	120

χ^2 – test (4)

Example (continued):

COMBINATION	ACTUAL NUMBER	EXPECTED NUMBER	DIFFERENCE
INCOME_HIGH and CREDIT_YES	50	33.3	16.7
INCOME_HIGH and CREDIT_NO	0	16.7	-16.7
INCOME_LOW and CREDIT_YES	30	46.7	-16.7
INCOME_LOW and CREDIT_NO	40	23.3	16.7

- ♦ the value of the χ^2 statistics: 42.857
- ♦ at the level of significance $\alpha = 0.05$, the value of the χ^2 distribution with one degree of freedom is equal to:

$$\chi^2_{(1)}(0.05) = 3.84$$

= > there is dependence between the income height and provision of credit

Fisher's – test

- χ^2 – test can be used only for big enough frequencies – when $(r_k \cdot s_l) / n \geq 5 \quad \forall k, l$
- For the four-field contingency tables, it is possible to use Fisher's test (applicable to low frequencies)
- Computation of the probability that the four-field table has actual frequencies a_{kl} with the given marginal frequencies r and s :

$$p = \frac{r_1! r_2! s_1! s_2!}{n! a_{11}! a_{12}! a_{21}! a_{22}!} \quad \left(= \frac{\binom{r_1}{a_{11}} \binom{r_2}{a_{21}}}{\binom{n}{s_1}} \right)$$

Fisher's – test (2)

- The probabilities p will be summed up for all considered values of actual frequencies at the given marginals (we assume $a_{11} = \min_{kl} a_{kl}$):

$$P = \sum_{i=0}^{a_{11}} \frac{r_1! \ r_2! \ s_1! \ s_2!}{n! \ (a_{11} - i)! \ (a_{12} + i)! \ (a_{21} + i)! \ (a_{22} - i)!}$$

- If $P \leq \alpha$, the null hypothesis will be rejected at the level of significance α

Fisher's – test (3)

Exercises

9.1 Of 30 men employed in a small workshop 18 worked in one department and 12 in another department. In one year five of the 18 reported sick with septic hands, and of the 12 men in the other department one did so. Is there a difference in the departments and how would you report this result?

	S+	S-	
DA	5	13	18
DB	1	11	12
	6	24	30

==>

P ₁	S+	S-	
DB	1	11	12
DA	5	13	18
	6	24	30

$$P = P_0 + P_1 = 0.204 > 0.05$$

⇒ **Accept the null hypothesis of independence.**

P ₀	S+	S-	
DB	0	12	12
DA	6	12	18
	6	24	30

$$P_0 = \frac{12!18!6!24!}{30!0!6!12!12!} = 0.031$$

$$P_1 = \frac{12!18!6!24!}{30!1!5!11!13!} = 0.173$$

Fisher's – test (4)

P ₂	S+	S-	
DB	2	10	12
DA	4	14	18
	6	24	30

P ₄	S+	S-	
DB	4	8	12
DA	2	16	18
	6	24	30

P ₆	S+	S-	
DB	6	6	12
DA	0	18	18
	6	24	30

P ₃	S+	S-	
DB	3	9	12
DA	3	15	18
	6	24	30

P ₅	S+	S-	
DB	5	7	12
DA	1	17	18
	6	24	30

$$P_2 = \frac{12!18!6!24!}{30!2!4!10!14!} = 0.340$$

$$P_3 = \frac{12!18!6!24!}{30!3!3!9!15!} = 0.302$$

$$P_4 = \frac{12!18!6!24!}{30!4!2!8!16!} = 0.128$$

$$P_5 = \frac{12!18!6!24!}{30!5!1!7!17!} = 0.024$$

$$P_6 = \frac{12!18!6!24!}{30!6!0!6!18!} = 0.002$$

Check:

$$\begin{aligned} \circ \sum_{i=0}^6 P_i &= 0.031 + 0.173 + 0.340 + 0.302 + \\ &\quad + 0.128 + 0.024 + 0.002 = 1.000 \end{aligned}$$

DATA ANALYSIS

Contents:

- *Data standardization techniques*
- *Range, quartiles, and outliers*
- **Relationship between the attributes**
- *Discriminant analysis*
- *Cluster analysis*

Contents:

- *Data standardization techniques*
- *Range, quartiles, and outliers*
- **Relationship between the attributes**
 - *Relationship between categorical variables*
 - χ^2 - test and Fisher's test
 - **Regression analysis (~ relationship between numerical variables)**
 - **Linear regression for one input variable and correlation coefficients**
 - **Multi-dimensional linear regression**
 - **Logistic regression**
- *Discriminant analysis*
- *Cluster analysis*
 - *The main clustering approaches*
 - *Distance measures*
 - *k-means clustering*
 - *Hierarchical clustering*
 - *Vector quantization*
 - *Other clustering techniques*
 - *k-medoids algorithm, grid-based methods, and density-based algorithms*
 - *Scalable data clustering approaches*

Regression analysis

~ **determine the relationship of variable Y to other (one or more) variables $X_1, \dots, X_m$**

Reasons for regression analysis:

1. It is very costly to obtain / measure the output values => predict the output based on the inputs that are easily available
2. Input values are known earlier than the output => estimate the output
3. Controlled input values might help find correct behavior of the output
4. There can be a causal relationship between the input and output of a system => find this relationship

Regression analysis (2)

- Correlation analysis

Is there a linear relationship between two numerical quantities?

- Linear regression

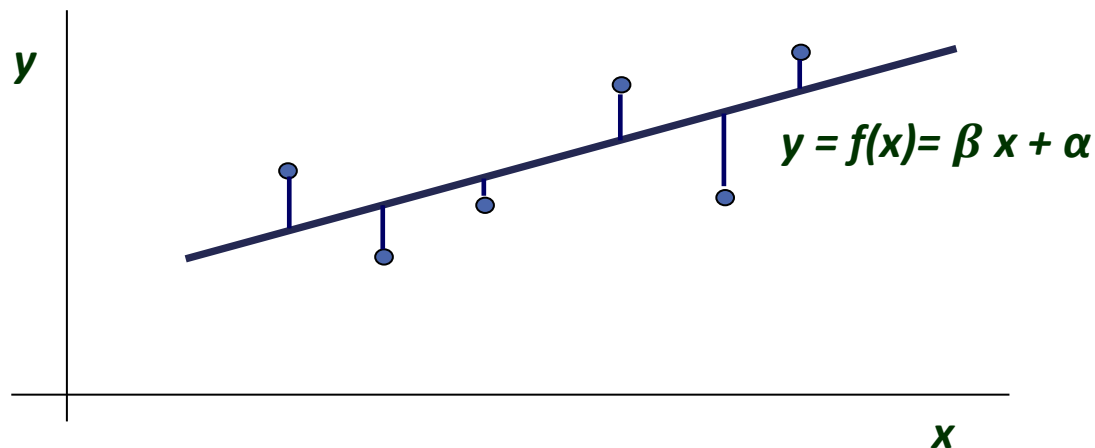
What are the parameters of the linear relationship between two numerical quantities?

→ Approximation of the observed values $[x_i, y_i]; i = 1, \dots$

by means of $y = \beta x + \alpha + \varepsilon$

Regression analysis (3)

→ **least squares method** (minimizes the squared differences between the actual and expected values)



→ look for: $\min \sum_{i=1}^n (y_i - f(x_i))^2$

Regression analysis (4)

Generalized regression equation:

$$Y = \alpha + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_m X_m$$

⇒ for the individual patterns:

$$y_i = \alpha + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_m x_{im} + \varepsilon_i$$

regression deviation for the pattern *i*



Regression analysis (5)

Linear regression for one input variable:

- Patterns: $(x_1, y_1), \dots, (x_n, y_n)$; $x_i \in X, y_i \in Y$
- Regression equation: $Y = \alpha + \beta X$



regression coefficients

- Use the least squares method to find regression coefficients!
- Quadratic error

$$SSE = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - y'_i)^2 = \sum_{i=1}^n (y_i - \alpha - \beta x_i)^2$$

Regression analysis (6)

$$SSE = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - y'_i)^2 = \sum_{i=1}^n (y_i - \alpha - \beta x_i)^2$$

- Derivative of the quadratic error with respect to α and β :

$$\frac{\partial(SSE)}{\partial \alpha} = -2 \sum_{i=1}^n (y_i - \alpha - \beta x_i)$$

$$\frac{\partial(SSE)}{\partial \beta} = -2 \sum_{i=1}^n ((y_i - \alpha - \beta x_i)x_i)$$

- Minimization of the error (the derivatives should be equal to 0)

Regression analysis (7)

$$\frac{\partial(SSE)}{\partial\alpha} = -2 \sum_{i=1}^n (y_i - \alpha - \beta x_i) \quad \frac{\partial(SSE)}{\partial\beta} = -2 \sum_{i=1}^n ((y_i - \alpha - \beta x_i)x_i)$$

$$n\alpha + \beta \sum_{i=1}^n x_i = \sum_{i=1}^n y_i$$

$$\alpha \sum_{i=1}^n x_i + \beta \sum_{i=1}^n x_i^2 = \sum_{i=1}^n x_i y_i$$

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

$$\bar{y} = \frac{\sum_{i=1}^n y_i}{n}$$

$$\alpha + \beta \bar{x} = \bar{y} \quad \text{hence} \quad \alpha = \bar{y} - \beta \bar{x}$$

$$(\bar{y} - \beta \bar{x}) \sum_{i=1}^n x_i + \beta \sum_{i=1}^n x_i^2 = \sum_{i=1}^n x_i y_i$$

Regression analysis (8)

$$\alpha = \bar{y} - \beta \bar{x} \quad (\bar{y} - \beta \bar{x}) \sum_{i=1}^n x_i + \beta \sum_{i=1}^n x_i^2 = \sum_{i=1}^n x_i y_i$$

$$\beta \left(\sum_{i=1}^n x_i^2 - \bar{x} \sum_{i=1}^n x_i \right) = \sum_{i=1}^n x_i y_i - \bar{y} \sum_{i=1}^n x_i$$

$$\beta \sum_{i=1}^n x_i (x_i - \bar{x}) = \sum_{i=1}^n x_i (y_i - \bar{y}) \quad \text{hence} \quad \beta = \frac{\sum_{i=1}^n x_i (y_i - \bar{y})}{\sum_{i=1}^n x_i (x_i - \bar{x})}$$

Predict y by means of $y = \alpha + \beta x$

Regression analysis (9)

$$\beta = \frac{\sum_{i=1}^n x_i (y_i - \bar{y})}{\sum_{i=1}^n x_i (x_i - \bar{x})}$$

◆ after re-writing: $\beta = \frac{\sum_{i=1}^n (x_i - \bar{x}) (y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$

$$\text{and } \alpha = \bar{y} - \beta \bar{x}$$

($\beta \sim$ sample covariance / sample variance in x)

Regression analysis (10)

Correlation coefficients:

An assessment of the „amount“ of linear dependence

$$\rho(x, y) = \frac{S_{xy}}{\sqrt{S_x^2 S_y^2}} \quad ; \quad \bar{x} = \frac{\sum_i x_i}{n} \quad ; \quad \bar{y} = \frac{\sum_i y_i}{n}$$

(corrected) sample covariance: $S_{xy} = \frac{1}{(n-1)} \sum_i (x_i - \bar{x})(y_i - \bar{y})$

(corrected) sample variances: $S_x^2 = \frac{1}{(n-1)} \sum_i (x_i - \bar{x})^2$

$$S_y^2 = \frac{1}{(n-1)} \sum_i (y_i - \bar{y})^2$$

Regression analysis - an example

	GDP per capita	Agric.	Nuc. electr.	Child. Out of school	Pupil / teach prim.	Pupil / teach sec- ond.	Pupil / teach ter- tiary	Fem. pupils %	Life ex- pect.	Phys. / (1000 peo- ple)	Fert. rate	Tlf. Subs.	Ind. Using Inter- net	Mob. subs.	Pop. Ages 0-14	Pop. Ages 15-64	Pop. Ages 65+	Pop. den- sity
Pop. density	0.38	-0.36	0.11	0.15	-0.16	-0.04	0.10	-0.10	0.23	0.12	-0.34	0.35	0.13	0.08	-0.31	0.24	0.34	1.00
Pop. Ages 65+	0.84	-0.64	0.64	-0.48	-0.67	-0.48	-0.31	-0.26	0.70	0.76	-0.76	<u>0.88</u>	0.52	0.32	<u>-0.92</u>	0.72	1.00	
Pop. Ages 15-64	0.56	-0.72	0.35	-0.47	-0.67	-0.36	-0.34	-0.18	<u>0.82</u>	0.68	<u>-0.95</u>	0.62	0.47	0.42	<u>-0.93</u>	1.00		
Pop. Ages 0-14	-0.75	0.73	-0.53	0.51	0.73	0.45	0.35	0.24	<u>-0.82</u>	-0.77	<u>0.93</u>	-0.80	-0.53	-0.40	1.00			
Mobile subs.	0.47	-0.28	0.08	-0.29	-0.29	-0.10	0.05	-0.09	0.46	0.14	-0.39	0.38	<u>0.86</u>	1.00				
Ind. Using In- ternet	0.70	-0.40	0.23	-0.29	-0.39	-0.18	-0.06	-0.19	0.55	0.29	-0.45	0.54	1.00					
Telephone. Subs.	<u>0.90</u>	-0.66	0.77	-0.45	-0.51	-0.37	-0.17	-0.25	0.67	0.54	-0.68	1.00						
Fertility rate	-0.59	0.74	-0.39	0.44	0.64	0.37	0.31	0.11	<u>-0.88</u>	-0.67	1.00							
Physicians / (1000 people)	0.48	-0.52	0.40	-0.43	-0.63	-0.53	-0.46	-0.10	0.63	1.00								
Life expectancy	0.64	-0.71	0.41	-0.43	-0.59	-0.48	-0.31	-0.01	1.00									
Female pupils %	-0.30	0.20	-0.21	0.38	0.24	0.12	0.09	1.00										
Pupil/teacher tertiary	-0.17	0.26	0.04	0.16	0.30	0.35	1.00											
Pupil/teacher secondary	-0.38	0.30	-0.30	0.31	0.73	1.00												
Pupil/teacher primary	-0.51	0.54	-0.27	0.38	1.00													
Child. Out of school	-0.36	0.40	-0.37	1.00														
Nuclear elect. Prod.	0.68	-0.41	1.00															
Agriculture	-0.63	1.00																
GDP per capita	1.00																	

Source: I. Padrón Peña: World Bank data analysis – Spanish speaking countries

Regression analysis (11)

Multi-dimensional regression:

- **Linear**

we assume a linear dependence of the explained (dependent) quantity y on several explaining (independent) quantities $x_1, x_2, \dots, x_m$

→ an assumption for the i -th observation:

$$y_i = \alpha + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_m x_{im} + \varepsilon_i$$

Regression analysis (12)

Multi-dimensional regression:

Linear (continued)

→ matrix notation: $\vec{y} = X \vec{\beta}$; $\vec{y} = (y_1, \dots, y_n)^T$; $\vec{\beta} = (\alpha, \beta_1, \dots, \beta_m)^T$

$$X = \begin{bmatrix} 1 & x_{11} & \dots & x_{1m} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \dots & x_{nm} \end{bmatrix}$$

→ solution of $\vec{y} = X \vec{\beta}$ by means of the least squares'

method: $\vec{\beta} = (X^T X)^{-1} X^T \vec{y}$

Regression analysis (13)

Multi-dimensional linear regression:

- variable Y is modeled as a linear function of several predictive variables

$$Y = \alpha + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_m X_m$$

- matrix notation

$$Y = X \vec{\beta} \quad \text{X ... extended matrix of input patterns}$$

Y ... matrix of the outputs

$$\vec{\beta} = (\alpha, \beta_1, \dots, \beta_m);$$

- quadratic deviation $SSE = (Y - X \vec{\beta})^T \cdot (Y - X \vec{\beta})$

Regression analysis (14)

- optimization step (LMS)

$$\frac{\partial(SSE)}{\partial \vec{\beta}} = \frac{\partial \left((Y - X \vec{\beta})^T (Y - X \vec{\beta}) \right)}{\partial \vec{\beta}} = 0$$

$$\Rightarrow (X^T \cdot X) \vec{\beta} = X^T \cdot Y$$

- regression coefficients are determined according to:

$$\vec{\beta} = (X^T \cdot X)^{-1} (X^T \cdot Y)$$

- Solution of complex practical tasks can be accompanied with high computational costs

$\Rightarrow$ approximative solutions

Regression analysis (15)

Multi-dimensional regression (continued):

□ **non-linear**

a more complex functional dependence is assumed between $\vec{x}$ and y – quadratic, exponential, ...

- **logistic regression** (an example for non-linear regression)
 - assume that the dependent variable y is categorical, e.g., a two-valued one
 - model the probability, that y has a certain value depending on the combination of the values of independent quantities $\vec{x}$
 - „conditional chance“: $P(y|\vec{x})/(1 - P(y|\vec{x}))$

Regression analysis (16)

Multi-dimensional regression (continued):

- **logistic regression (continued)**

- For **y** with binary values (**1** or **0**): $\frac{P(y=1|x_1, x_2, \dots, x_m)}{1-P(y=1|x_1, x_2, \dots, x_m)} = e^{\alpha + \sum_j \beta_j x_j}$

$$\ln \frac{P(y = 1|x_1, x_2, \dots, x_m)}{1 - P(y = 1|x_1, x_2, \dots, x_m)} = \alpha + \beta_1 x_1 + \dots + \beta_m x_m$$

resp.

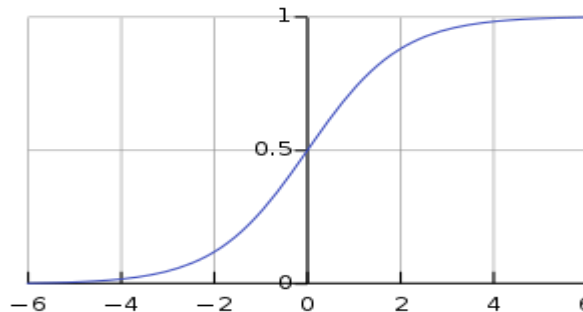
$$P(y = 1|x_1, x_2, \dots, x_m) = \frac{e^{\alpha + \sum_j \beta_j x_j}}{1 + e^{\alpha + \sum_j \beta_j x_j}} = \frac{1}{1 + e^{-\alpha - \sum_j \beta_j x_j}}$$

Regression analysis (17)

Multi-dimensional regression (continued):

- **logistic regression** (continued)
 - estimate the „chance“ / probability of **$y = 1$** :

the sigmoid function:



$$\frac{1}{(1 + \exp(-sum))}$$
$$sum = \alpha + \sum_j \beta_j x_j$$

- estimate the parameters of the model by means of the **maximum likelihood method** (maximization of **L**):

$$L = \prod_{i=1}^n P(y_i | x_{i,1}, x_{i,2}, \dots, x_{i,m}) = \prod_{i=1}^n \left[\left(\frac{1}{1 + e^{-\alpha - \sum_j \beta_j x_{i,j}}} \right)^{y_i} \cdot \left(\frac{1}{1 + e^{\alpha + \sum_j \beta_j x_{i,j}}} \right)^{1-y_i} \right]$$

$$sum_i = \alpha + \sum_j \beta_j x_{i,j}$$

Regression analysis (18)

Multi-dimensional regression - logistic regression (continued):

- Instead of maximizing L , **minimize the cross-entropy loss $E = -\log L$** :

$$E(\vec{\beta} | X) = - \sum_i \left[y_i \log \left(\frac{1}{1 + e^{-\alpha - \sum_j \beta_j x_{i,j}}} \right) + (1 - y_i) \log \left(\frac{1}{1 + e^{\alpha + \sum_j \beta_j x_{i,j}}} \right) \right]$$

- The derivative of the sigmoid is: $s'(sum) = s(sum) (1 - s(sum))$
- Then, $E(\vec{\beta} | X) = - \sum_i [y_i \log s(sum_i) + (1 - y_i) \log (1 - s(sum_i))]$
- And for the adjustment rules, it follows that:

$$\begin{aligned} \Delta \beta_j &= -\eta \frac{\partial E}{\partial \beta_j} = \eta \sum_i \left(\frac{y_i}{s(sum_i)} - \frac{1 - y_i}{1 - s(sum_i)} \right) s(sum_i) (1 - s(sum_i)) x_{ij} \\ &= \eta \sum_i (y_i - s(sum_i)) x_{ij} \end{aligned}$$

$$\Delta \alpha = -\eta \frac{\partial E}{\partial \alpha} = \eta \sum_i (y_i - s(sum_i))$$

η denotes the learning rates

DATA ANALYSIS

Contents:

- *Data standardization techniques*
- *Range, quartiles, and outliers*
- *Relationship between the attributes*
- **Discriminant analysis**
- *Cluster analysis*

Contents:

- *Data standardization techniques*
- *Range, quartiles, and outliers*
- *Relationship between the attributes*
 - *Relationship between categorical variables*
 - χ^2 - test and Fisher's test
 - *Regression analysis (~ relationship between numerical variables)*
 - *Linear regression for one input variable and correlation coefficients*
 - *Multi-dimensional linear regression*
 - *Logistic regression*
- **Discriminant analysis**
- *Cluster analysis*
 - *The main clustering approaches*
 - *Distance measures*
 - *k-means clustering*
 - *Hierarchical clustering*
 - *Vector quantization*
 - *Other clustering techniques*
 - *k-medoids algorithm, grid-based methods, and density-based algorithms*
 - *Scalable data clustering approaches*

Discriminant analysis

~ classification of patterns into known classes

→ search for the dependence of the nominal quantity (that impacts the class membership) on the other ***m*** numerical quantities

- Assume, that to every class (~ value of the nominal quantity) ***c_t***; ***t = 1, ..., T*** there exists a (**discriminative**) **function *f_t***

$$f_t(\vec{x}) = \max_k f_k(\vec{x}) \quad ; \quad k = 1, \dots, T$$
$$\Leftrightarrow \vec{x} = (x_1, x_2, \dots, x_m) \text{ belongs to } c_t$$

Discriminant analysis (2)

Linear discriminant analysis:

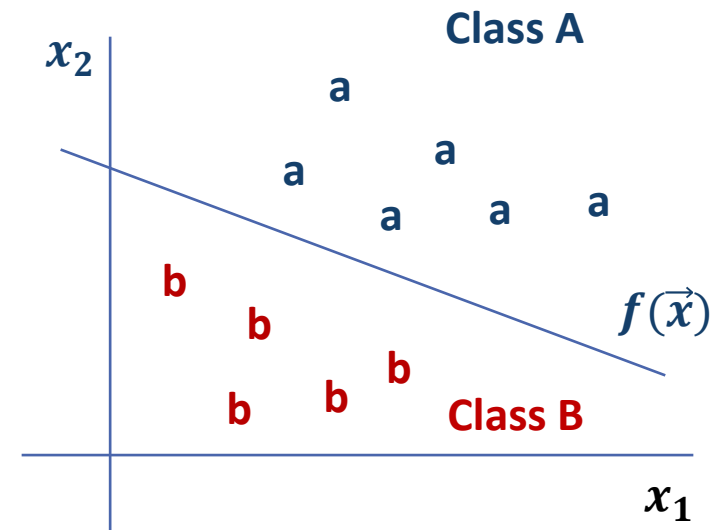
$$f_t = q_{0t} + q_{1t}x_1 + q_{2t}x_2 + \cdots + q_{mt}x_m$$

Discrimination into two classes:

- Instead of the functions f_1 and f_2 we can look for the function

$$f(\vec{x}) = f_1(\vec{x}) - f_2(\vec{x})$$

- Patterns are classified according to the sign of $f(\vec{x})$
 - “+” indicates class 1; “−” class 2



Discriminant analysis (3)

An optimum classification in the sense of a minimum error

→ discriminant function $\approx$ conditional (a posteriori) $\vec{x}$ probabilities of classifying observation $\vec{x}$ to the class $\mathbf{c}_t$

$$f_t(\vec{x}) = P(c_t|\vec{x}) = \frac{P(\vec{x}|c_t) \cdot P(c_t)}{\sum_k P(\vec{x}|c_k)P(c_k)}$$

→ for two classes:

$$\begin{aligned} f(\vec{x}) &= f_1(\vec{x}) - f_2(\vec{x}) = \\ &= P(\vec{x}|c_1) \cdot P(c_1) - P(\vec{x}|c_2) \cdot P(c_2) \end{aligned}$$

Discriminant analysis (4)

→ normal distribution (quadratic discriminant function) with the means $\vec{\mu}_1, \vec{\mu}_2$ and covariance matrices $\mathbf{S}_1$ and $\mathbf{S}_2$:

$$f(\vec{x}) = \frac{1}{2} \vec{x}^T (\mathbf{S}_1^{-1} - \mathbf{S}_2^{-1}) \vec{x} + (\vec{\mu}_1^T \mathbf{S}_1^{-1} - \vec{\mu}_2^T \mathbf{S}_2^{-1}) \vec{x} + \\ + \frac{1}{2} \ln \frac{|\mathbf{S}_2|}{|\mathbf{S}_1|} - \frac{1}{2} (\vec{\mu}_1^T \mathbf{S}_1^{-1} \vec{\mu}_1 - \vec{\mu}_2^T \mathbf{S}_2^{-1} \vec{\mu}_2) - \ln \frac{P(c_1)}{P(c_2)}$$

→ different means $\vec{\mu}_1$ and $\vec{\mu}_2$, identical covariance matrices: ($\mathbf{S}_1 = \mathbf{S}_2 = \mathbf{S}$, linear discriminant function)

$$f(\vec{x}) = (\vec{\mu}_1^T - \vec{\mu}_2^T) \mathbf{S}^{-1} \vec{x} - \frac{1}{2} (\vec{\mu}_1^T - \vec{\mu}_2^T) \mathbf{S}^{-1} \cdot (\vec{\mu}_1 - \vec{\mu}_2) - \ln \frac{P(c_1)}{P(c_2)}$$

Discriminant analysis (5)

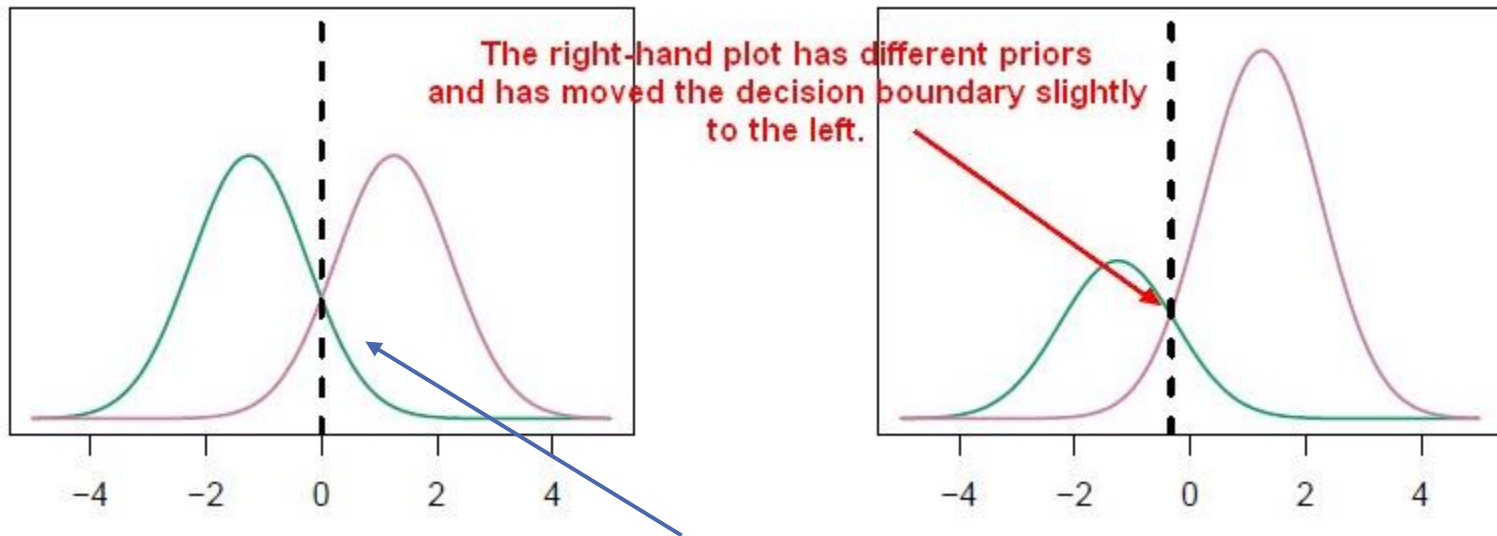
→ identity covariance matrices, both classes are equally probable

$$f(\vec{x}) = (\vec{\mu}_1^T - \vec{\mu}_2^T) \vec{x} - \frac{1}{2} (\vec{\mu}_1^T - \vec{\mu}_2^T) (\vec{\mu}_1 - \vec{\mu}_2)$$

→ for normal distributions, the search for discriminant functions will be reduced to the estimation of the means $\vec{\mu}_i$ (based on the sample means) and covariance matrices $\mathbf{S}_i$ (composed of corrected sample variances)

Discriminant analysis (6)

Examples:



**Normal distributions with the same variances
allow for discrimination according to
the estimated mean values**

DATA ANALYSIS

Contents:

- *Data standardization techniques*
- *Range, quartiles, and outliers*
- *Relationship between the attributes*
- *Discriminant analysis*
- **Cluster analysis**

Contents:

- *Data standardization techniques*
- *Range, quartiles, and outliers*
- *Relationship between the attributes*
 - *Relationship between categorical variables*
 - χ^2 - test and Fisher's test
 - *Regression analysis (~ relationship between numerical variables)*
 - *Linear regression for one input variable and correlation coefficients*
 - *Multi-dimensional linear regression*
 - *Logistic regression*
- *Discriminant analysis*
- **Cluster analysis**
 - **The main clustering approaches**
 - **Distance measures**
 - **k-means clustering**
 - **Hierarchical clustering**
 - **Vector quantization**
 - *Other clustering techniques*
 - *k-medoids algorithm, grid-based methods, and density-based algorithms*
 - *Scalable data clustering approaches*

Cluster analysis

- Divide the observed patterns into groups (clusters) of mutually similar patterns?

→ Assumption: we can measure the distance between patterns

- Each pattern is characterized by ***m*** numerical quantities
- Distance between two patterns:

$$\vec{x}_1 = (x_{11}, \dots, x_{1m}) \text{ and } \vec{x}_2 = (x_{21}, \dots, x_{2m})$$

Cluster analysis (2)

1. Hamming distance: $d_H(\vec{x}_1, \vec{x}_2) = \sum_{j=1}^m |x_{1j} - x_{2j}|$

(~ Manhattan)

2. Euclidean distance: $d_E(\vec{x}_1, \vec{x}_2) = \sqrt{\sum_{j=1}^m (x_{1j} - x_{2j})^2}$

3. Chebyshev distance: $d_C(\vec{x}_1, \vec{x}_2) = \max_j |x_{1j} - x_{2j}|$

4. Minkovski metrics (1.-3. are its special cases):

$$L^{(z)}(\vec{x}_1, \vec{x}_2) = \sqrt[z]{\sum_{j=1}^m (x_{1j} - x_{2j})^z}$$

Cluster analysis (3)

$$d_H(\vec{x}_1, \vec{x}_2) = L^{(1)}(\vec{x}_1, \vec{x}_2)$$

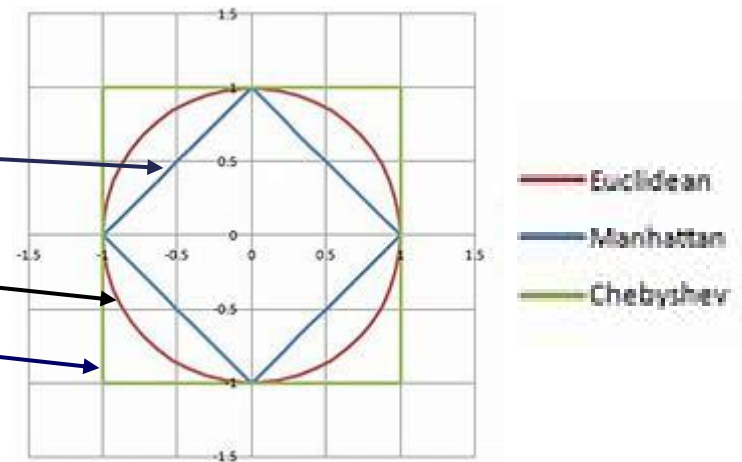
$$d_E(\vec{x}_1, \vec{x}_2) = L^{(2)}(\vec{x}_1, \vec{x}_2)$$

$$d_C(\vec{x}_1, \vec{x}_2) = \lim_{z \rightarrow \infty} L^{(z)}(\vec{x}_1, \vec{x}_2)$$

$$d_H(0, \vec{x}_2) = 1$$

$$d_E(0, \vec{x}_2) = 1$$

$$d_C(0, \vec{x}_2) = 1$$



Cluster analysis (4)

- The choice of the distance metrics depends on the involved quantities and their range
 - normalize the quantities (~ divide them by their mean, standard deviation, range (max – min), ...)
- We assume the same variance for all the quantities
- Different variance for the quantities
 - **Mahalanobis distance:**

$$d_{M^2}(\vec{x}_1, \vec{x}_2) = (\vec{x}_1 - \vec{x}_2)^T S^{-1} (\vec{x}_1 - \vec{x}_2)$$

S – the co-variance matrix

Cluster analysis (5)

The distance between two clusters U and V :

- the method of the nearest neighbor
 - ~ the minimum from the distances between their elements

$$D(U, V) = \min_{k,l} d(\vec{x}_k, \vec{x}_l) \quad ; \quad \vec{x}_k \in U, \quad \vec{x}_l \in V$$

- the method of the farthest neighbor
 - ~ the maximum from the distances between their elements

$$D(U, V) = \max_{k,l} d(\vec{x}_k, \vec{x}_l) \quad ; \quad \vec{x}_k \in U, \quad \vec{x}_l \in V$$

Cluster analysis (6)

The distance between two clusters U and V (continued):

- The method of average distance $\sim$ the average of the distances between the patterns; ($n_U \sim$ number of patterns in cluster U ; $n_V \sim$ number of patterns in cluster V)

$$D(U, V) = \frac{1}{n_U n_V} \sum_{k=1}^{n_U} \sum_{l=1}^{n_V} d(\vec{x}_k, \vec{x}_l) \quad ; \quad \vec{x}_k \in U, \quad \vec{x}_l \in V$$

- Centroid method $\sim$ the distance between the centers of the clusters; ($\vec{v} \sim$ the center of the cluster V ; $\vec{u} \sim$ the center of the cluster U)

$$D(U, V) = d(\vec{u}, \vec{v})$$



https://www.idnes.cz/revue/ankety/PA20110218_nh_572

Cluster analysis (7)

Centroid ~ the center of the cluster

- A „prototype“ that represents the given cluster
- The same cluster can be represented by one or even more centroids at the same time
 - depending on the shape of the cluster and the metrics chosen to evaluate the distances
- Clustering by means of the *k*-means method



Source: https://www.huffingtonpost.ca/2013/10/07/average-woman-face-around-world_n_4058145.html

A toy application to test: <http://faceresearch.org/demos/average>



Cluster analysis (8)

Clustering by means of the k -means method (Lloyd, 1982):

1. Divide the patterns randomly into k clusters U_k ; $U_k \neq \emptyset$, $\bigcap_{k=1}^K U_k = \emptyset$, $1 \leq k \leq K$
2. Determine the centroids of all the clusters in recent division,

e.g., according to: $\vec{c}_k = \frac{1}{|U_k|} \sum_{\vec{x}_i \in U_k} \vec{x}_i$

3. For each pattern $\vec{x}$
 1. Determine the distance $d(\vec{x}, \vec{c}_k)$; $1 \leq k \leq K$; $\vec{c}_k$ ~ the centroid of the k -th cluster)
 2. Let $d(\vec{x}, \vec{c}_l) = \min_k d(\vec{x}, \vec{c}_k)$
 3. If $\vec{x}$ does not belong to cluster l (with the closest distance to centroid $\vec{c}_l$),
move $\vec{x}$ to cluster l
4. If there was a transfer, then goto Step 2, else FINISH

Cluster analysis (8a)

Clustering by means of the k -means method:

- Variants of the algorithm:
 - At the initial division, pronounce the first k patterns to be centroids (eliminates Step 2)
 - Adjustment of the centroids after every transfer (in the cycle of Step 3)
- Afterwards, the clusters are represented by their centroids

Cluster analysis (8b)

Extensions to the Lloyd algorithm:

- alternating minimization steps do not provide approximation guarantee, yet the Lloyd's algorithm never increases the cost of the initial clustering
=> provide the algorithm with a provably good initialization
- **k-means++ seeding** (Arthur and Vassilvitskii 2007):
 - Sample **k** of the **P** patterns with probability $d^2(\vec{x}, \vec{c}_l) / \sum_{q=1}^P d^2(\vec{x}_q, \vec{c}_l)$ proportional to their squared distance to the closest existing centroid $\vec{c}_l$
- **LocalSearch++** (Lattanzi and Sohler, 2019):
 - First, use k-means++ to pick **k** initial centroids
 - Afterwards, new patterns are sampled with an analogue probability. Sampled patterns replace existing centroids if they decrease the overall squared distance between the patterns and their closest centroids.

Cluster analysis (9)

Clustering by means of the k -medians algorithm:

- Manhattan distance is used as the objective function, hence, the distance $d(\vec{x}_i, \vec{y}_j)$ is defined as: $d(\vec{x}_i, \vec{y}_j) = \|\vec{x}_i - \vec{y}_j\|_1$.
- it can be shown that this way, the optimal representative $\vec{y}_j$ is the median of the data points along each dimension in cluster C_j .
- As the median is chosen independently along each dimension, the resulting m -dimensional representative will (typically) not belong to the original data set D .
- In general, the k -medians approach selects the cluster representatives in a more robust way than k -means, because the median is not so sensitive to the presence of outliers as the mean

Cluster analysis (10)

Hierarchical clustering algorithm:

~ a „bottom up“ approach

Initialization:

1. Determine mutual distances between all patterns
2. Assign each pattern to an individual cluster

Main cycle:

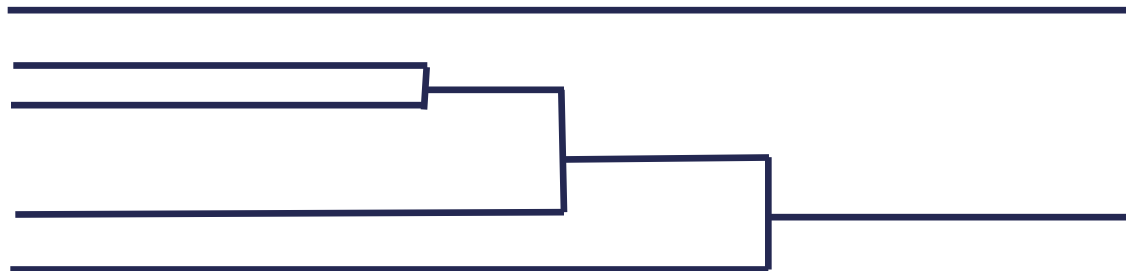
1. While there are more clusters (than one)
 1. Find two mutually closest clusters and merge them together
 2. For the new cluster, compute its distance from other clusters

Cluster analysis (11)

Hierarchical clustering algorithm (continued):

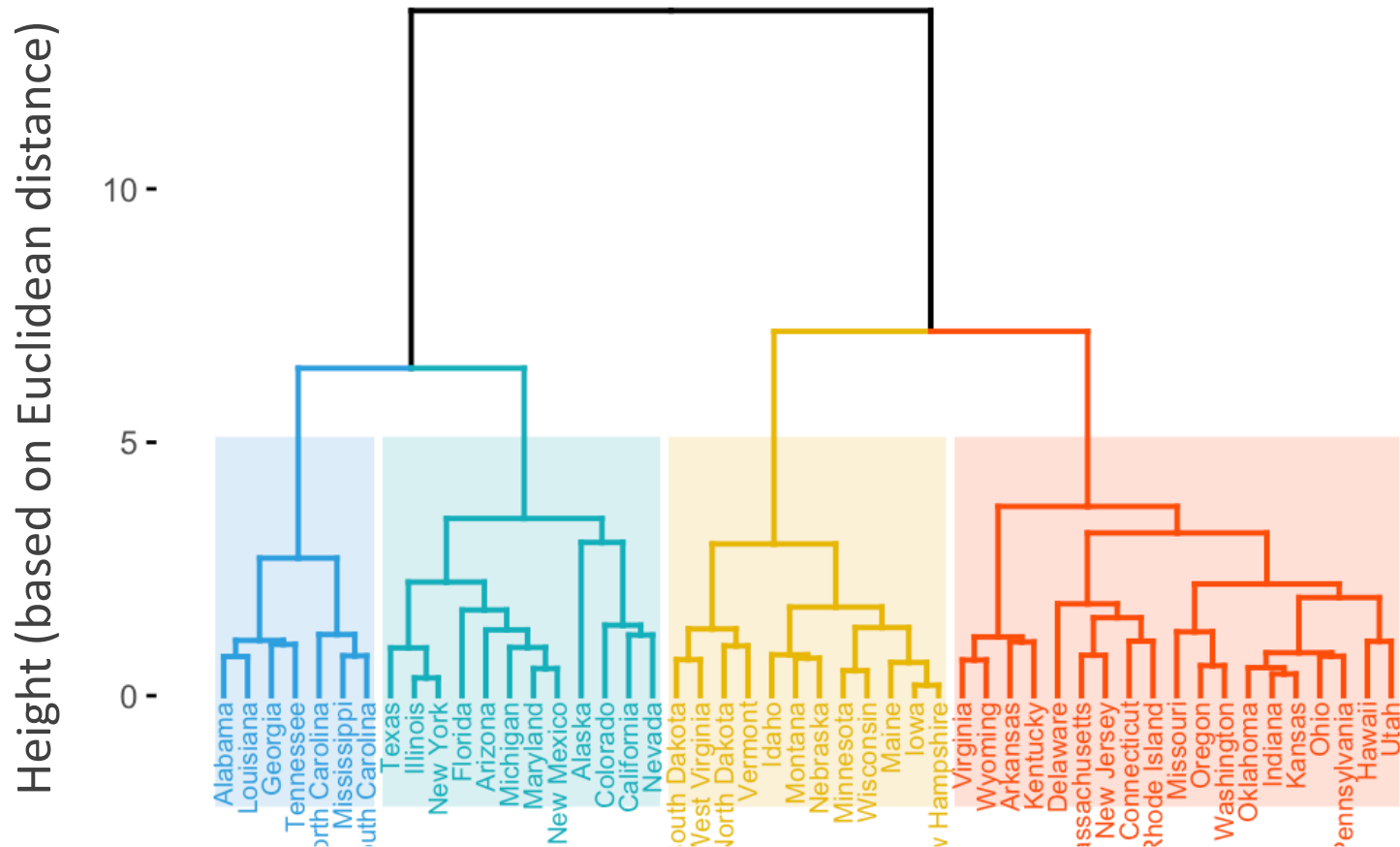
dendrogram

- ~ shows (from the left to the right) gradual merging of clusters
- ~ the optimum number of clusters is not known in advance



Cluster analysis (11a)

An example dendrogram for the USArrests data ([USArrests \(kaggle.com\)](https://www.kaggle.com/datasets/arslaner/USArrests))



Vector Quantization: the LVQ – Algorithm

for supervised clustering, classes are denoted by C

Step 1: Initialize all weight vectors $\vec{w}_j(0)$

Initialize the learning rates $\mu(0)$ and set $k = 0$

Step 2: Test the stop condition:

IF FALSE $\Rightarrow$ CONTINUE

IF TRUE $\Rightarrow$ QUIT

Step 3: For each training pattern $\vec{x}_i$ perform Step 4 and 5

Step 4: Determine the index of the weight vector ($j = q$) such that

$$\min_j \|\vec{x}_i - \vec{w}_j(k)\|_2^2$$

(use the Euclidean distance, $\vec{w}_q(k)$ is the weight vector with the minimum distance

Vector Quantization: the LVQ – Algorithm (2)

for supervised clustering, classes are denoted by C

Step 5: Adjust the respective weight vector $\vec{w}_q(k)$ according to:

$$\text{if } C_{\vec{w}_q} = C_{\vec{x}_i} \Rightarrow \vec{w}_q(k+1) = \vec{w}_q(k) + \mu(k)[\vec{x}_i - \vec{w}_q(k)]$$

$$\text{if } C_{\vec{w}_q} \neq C_{\vec{x}_i} \Rightarrow \vec{w}_q(k+1) = \vec{w}_q(k) - \mu(k)[\vec{x}_i - \vec{w}_q(k)]$$

Step 6: Set $k \leftarrow k + 1$

Decrease the learning rates, e.g., according to:

$$\mu(k) = \mu(k-1) / (k+1) \quad (k > 0)$$

Goto **Step 2**

Vector Quantization: the LVQ – Algorithm (3)

MATLAB: The function for LVQ1 (~ a variant of LVQ)

```
Function W = lvq1(X,CX,m,mu,maxiter)
% W = lvq1(X,CX,m,mu,maxiter) computes the weight matrix
%      for vector quantization LVQ1
% X:   is the input matrix (each column corresponds to an
%      input vector
% CX:  is a row vector of „scalar“ classes that correspond
%      to the column vectors from X
% m:   the number of (different) classes
% mu:  initial learning rates
% maxiter: the maximum number of iterations
N = size(X,2);    (~ the number of patterns, i.e., columns of X)
```

Vector Quantization: the LVQ – Algorithm (4)

MATLAB: The function for LVQ1 (continued)

```
% the weight vectors are initialized by the first m vectors
% from the training set (must include patterns from all the classes)
% this process automatically calibrates the neurons with the classes
W = X(:,1:m);
CW = CX(1:m);          % the classes for the weight vectors
snorm = zeros(1,m);
niter = 1;
while niter <= maxiter
    if niter == 1
        for i = m+1:N
            for j = 1:m
                snorm(1,j) = norm(X(:,i) - W(:,j))^2;
            end
            [mind,index] = min(snorm);
        end
    end
    niter = niter + 1;
end
```

Vector Quantization: the LVQ – Algorithm (5)

MATLAB: The function for LVQ1 (continued)

```
        if CX(i) == CW(index)
            W(:,index) = W(:,index) +
                        mu*(X(:,i)-W(:,index));
        else
            W(:,index) = W(:,index) -
                        mu*(X(:,i)-W(:,index));
        end
    end
else
    for i = 1:N
        for j = 1:m
            snorm(1,j) = norm(X(:,i)-W(:,j))^2;
        end
        mind,index] = min(snorm);
```

Vector Quantization: the LVQ – Algorithm (6)

MATLAB: The function for LVQ1 (continued)

```
        if CX(i) == CW(index)
            W(:,index) = W(:,index) + (mu/niter)*
                           (X(:,i)-W(:,index));
        else
            W(:,index) = W(:,index) - (mu/niter)*
                           (X(:,i)-W(:,index));
        end
    end
end
niter = niter + 1;
end
```

DATA ANALYSIS

Contents:

- *Data standardization techniques*
- *Range, quartiles, and outliers*
- *Relationship between the attributes*
- *Discriminant analysis*
- **Cluster analysis**

Contents:

- *Data standardization techniques*
- *Range, quartiles, and outliers*
- *Relationship between the attributes*
 - *Relationship between categorical variables*
 - χ^2 - test and Fisher's test
 - *Regression analysis (~ relationship between numerical variables)*
 - *Linear regression for one input variable and correlation coefficients*
 - *Multi-dimensional linear regression*
 - *Logistic regression*
- *Discriminant analysis*
- **Cluster analysis**
 - *The main clustering approaches*
 - *Distance measures*
 - *k-means clustering*
 - *Hierarchical clustering*
 - *Vector quantization*
 - **Other clustering techniques**
 - **k-medoids algorithm**, grid-based methods, and density-based algorithms
 - **Scalable data clustering approaches**

Variants of the techniques applicable to cluster analysis

The k -medoids algorithm:

- considers a data set $\mathcal{D}$ consisting of n d -dimensional data points $\bar{X}_1 \dots \bar{X}_n$.
- **The goal** is to determine k representatives $\bar{Y}_1 \dots \bar{Y}_k$ (**out of the original database $\mathcal{D}$** ; k , is to be specified by the user) that would minimize the objective function $\mathcal{O}$ given by the sum of the distances between the different data points and their closest representatives

Algorithm *GenericMedoids*(Database: $\mathcal{D}$, Number of Representatives: k)

begin

Initialize representative set S by selecting from $\mathcal{D}$;

repeat

Create clusters $(\mathcal{C}_1 \dots \mathcal{C}_k)$ by assigning
each point in $\mathcal{D}$ to closest representative in S
using the distance function $Dist(\cdot, \cdot)$;

Determine a pair $\bar{X}_i \in \mathcal{D}$ and $\bar{Y}_j \in S$ such that
replacing $\bar{Y}_j \in S$ with $\bar{X}_i$ leads to the
greatest possible improvement in objective function;

Perform the exchange between $\bar{X}_i$ and $\bar{Y}_j$ only
if improvement is positive;

until no improvement in current iteration;

return $(\mathcal{C}_1 \dots \mathcal{C}_k)$;

end

The objective function $\mathcal{O}$:

$$\mathcal{O} = \sum_{i=1}^n \left[\min_j Dist(\bar{X}_i, \bar{Y}_j) \right]$$

- time complexity of each iteration is $O(k \cdot n \cdot d)$; usually, a small constant number of iterations is necessary

Variants of the techniques applicable to cluster analysis (2)

The k -medoids algorithm - Why select the representatives from D ?

1. k -means clusters may be distorted by outliers
 - => cluster representatives can be located in empty regions, unrepresentative of most of the data points in that cluster
 - => such representatives may lead to partial merging of different clusters
 - Yet this problem can be resolved (at least partially) with careful outlier handling
 2. it might be difficult to compute the optimal central representative for complex data, e.g., a set of time series of varying length
 - => selecting representatives from the original dataset may be very helpful.
- ⇒ **the k -medoids algorithm can be defined virtually for any data type, as long as an appropriate similarity or distance function can be provided**

Variants of the techniques applicable to cluster analysis (3)

The k -medoids algorithm:

- uses hill-climbing with the representative set S initialized to a set of points from D
- S is iteratively improved by exchanging a point from S with a data point from D
- Possible strategies for the exchange:
 1. try all $|S| \cdot |D|$ possibilities for replacing a representative in S with a data point from D and then select the best one => **extremely expensive!**
 2. use a randomly selected set of r pairs $(\bar{X}_i, \bar{Y}_j)$ for a possible exchange ($\bar{X}_i$ is from D , $\bar{Y}_j$ is from the representative set S); the best of the r pairs is then used for the exchange.

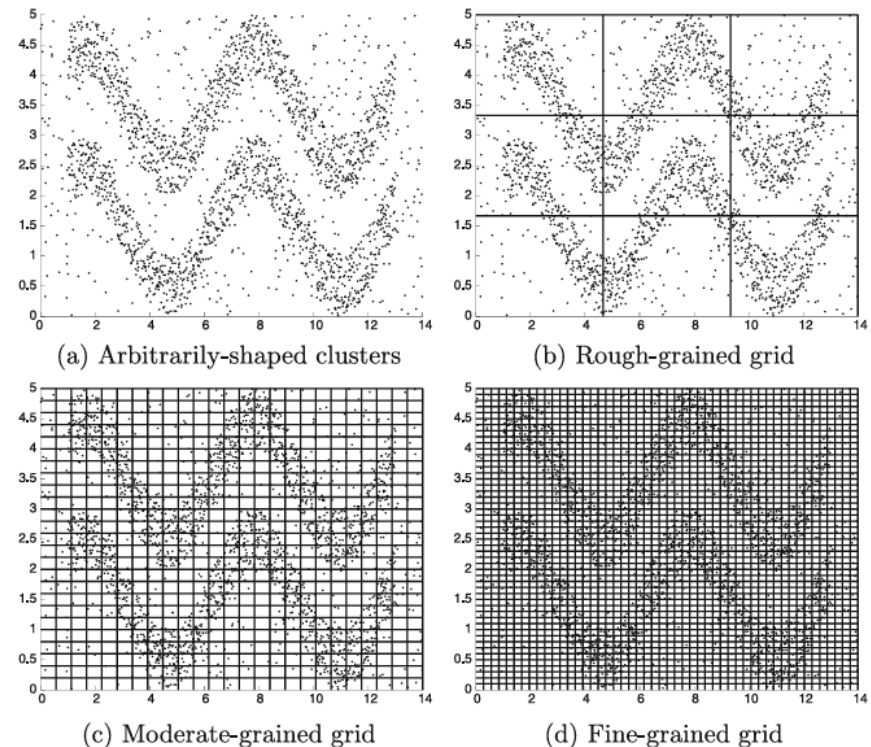
This solution requires time proportional to $r \cdot |D|$ and is said to have converged when the objective function does not improve, or if the average objective function improvement is below a user-specified threshold in the previous iteration.

- slower than the k -means method but applicable to various data types
- a scalable version represents the CLARA/CLARANS algorithm.

Variants of the techniques applicable to cluster analysis (4)

Grid-based methods (e.g., MAFIA):

- Discretize the data into p (equi-width) intervals yielding p^m hyper-cubes for an m -dimensional dataset.
- The resulting **hyper-cubes** are the building blocks for the following clustering.
- A density threshold τ is used to identify **dense** hyper-cubes (with at least τ points).
- An arbitrarily shaped cluster involves multiple dense regions that are **connected** together (two grid regions are connected if they share a side).
- **The objective:** determine the clusters as dense mutually connected regions



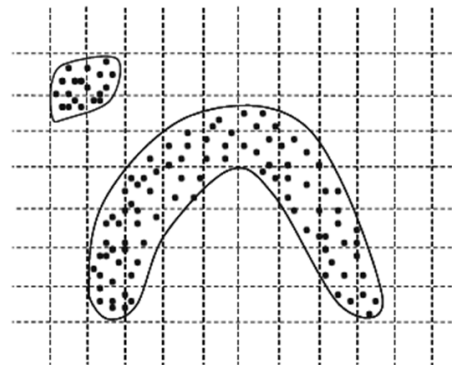
Clusters of arbitrary shape and grid partitions of different granularity

Variants of the techniques applicable to cluster analysis (5)

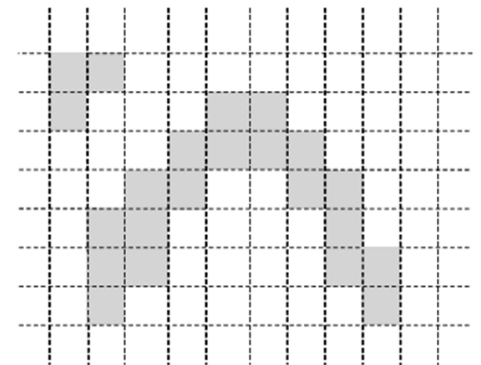
Grid-based methods:

- Dense grid cells are associated with nodes in a graph-based model used for clustering. The edges stand for adjacent connectivity.
- The data covered by the cells from the connected components of the graph form the final clusters

Agglomerating adjacent grids



(a) Data points and grid



(b) Agglomerating adjacent grids

Algorithm *GenericGrid*(Data: $\mathcal{D}$, Ranges: p , Density: τ)

begin

Discretize each dimension of data $\mathcal{D}$ into p ranges;

Determine dense grid cells at density level τ ;

Create graph in which dense grids are connected if they are adjacent;

Determine connected components of graph;

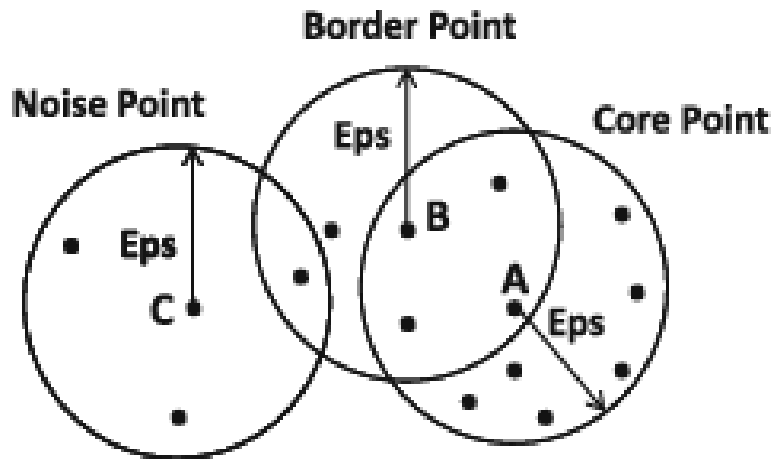
return points in each connected component as a cluster;

end

Variants of the techniques applicable to cluster analysis (6)

Density-based algorithms, e.g., DBSCAN, DENCLUE:

- Density characteristics of data points are used to merge them into clusters.
- Instead of grid cells, individual data points, resp. circular regions around them, form the building blocks. Their density is given by the number of points that lie within their radius Eps (including the point itself) – **core**, **border**, or **noise points**.



1. **Core point:** if the spherical region of Eps around it contains at least τ data points.
2. **Border point:** if the spherical region of Eps around it contains less than τ points, but also at least one core point within Eps .
3. **Noise point:** a data point that is neither a core point nor a border point.

Variants of the techniques applicable to cluster analysis (6)

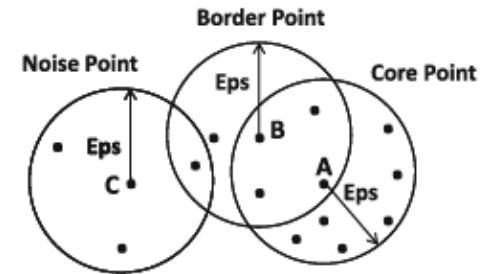
DBSCAN:

~ Density-Based Spatial Clustering of Applications with Noise

- Determine the **core**, **border**, and **noise** points
- Construct a **connectivity graph**, in which each node corresponds to a core point; an edge is added between a pair of core points, if and only if they lie within the distance of Eps from one another.
- Identify all **connected components** of the connectivity graph. They correspond to the clusters (constructed on the core points).
 - The border points are then assigned to the cluster with which they have the highest level of connectivity.
 - The resulting groups are reported as **clusters**.
 - The noise points are reported as **outliers**.

Variants of the techniques applicable to cluster analysis (6)

DBSCAN: Density-Based Spatial Clustering of Applications with Noise



Algorithm *DBSCAN*(Data: $\mathcal{D}$, Radius: Eps , Density: τ)

begin

Determine core, border and noise points of $\mathcal{D}$ at level (Eps, τ) ;

Create graph in which core points are connected

if they are within Eps of one another;

Determine connected components in graph;

Assign each border point to connected component

with which it is best connected;

return points in each connected component as a cluster;

end

Scalable data clustering

The CLARA/CLARANS approach:

- in many clustering applications, the size of the data is very large
 - the data cannot be stored in main memory and needs to reside on disk
 - a challenge with constraints on the algorithmic design of the methods
- *CLARA and CLARANS* are scalable implementations of *k*-medoids
 - inherit both the advantages and disadvantages of the base algorithm
 - e.g., the **CLARANS** algorithm is **easily generalizable to different data types** (beyond numeric data), but it inherits the relatively **high computational complexity** of *k*-medoids
- The choice of the most suitable algorithm depends on the application at hand!

Scalable data clustering (2)

The generic k -medoids approach:

- works with a set of representatives
- iteratively attempts to exchange a medoid with a non-medoid to improve the clustering quality
- allows flexibility in deciding how to execute the exchange

The CLARA method (Clustering LARge Applications):

- based on a variant of the k -medoids method - Partitioning Around Medoids (PAM)
- all possible $k \cdot (n-k)$ pairs of medoids and non-medoids are tested for an exchange
- the pair with the best improvement of the objective function is then exchanged
- Exchanges are performed until the algorithm converges to a locally optimal value
- for each iteration, the exchange process requires $O(k \cdot n^2)$ distance computations, i.e., $O(k \cdot n^2 \cdot d)$ time for a d -dimensional data set

=> RATHER EXPENSIVE!

Scalable data clustering (3)

The CLARA method - continued:

- Complexity can be reduced by applying the algorithm to a smaller sample set of size $f \cdot n$ to find the medoids (f is a sampling fraction, $f \ll 1$)
- The remaining (non-sampled) data points are assigned to the medoids found in the smaller sample.
- The above approach is applied repeatedly over independently chosen samples of data points of the same size $f \cdot n$.
- The best clustering over all these independently chosen samples is then selected as the optimal solution.
- For small values of f , CLARA can be orders of magnitude faster
 - the complexity of each iteration is then: $O(k \cdot f^2 \cdot n^2 \cdot d + k \cdot (n-k))$
- **The main problem with CLARA occurs when the preselected samples do not include a good choice of the medoids!**

Scalable data clustering (4)

The CLARANS approach:

~ Clustering Large Applications based on RANdomized Search

- works with the full data to avoid the problem with incorrectly preselected samples
- iteratively tests exchanges between random medoids with random non-medoids.
 - after each trial, it is checked if the quality improves.
 - if the quality improves, the attempted exchange is made final
 - otherwise, the number of unsuccessful exchange attempts is counted
 - a local optimal solution is said to have been found when a user-specified number of unsuccessful attempts *MaxAttempt* has been reached.
- the entire process of finding the local optimum is repeated for a user-specified number of iterations, denoted by *MaxLocal*.
- The clustering objective function of each of these *MaxLocal* locally optimal solutions is evaluated.
- The best among these local optima is selected as the optimal solution.

CLARANS explores a greater diversity of the search space than CLARA !

Scalable data clustering (5)

The CURE approach:

~ **Clustering Using Representatives** - an agglomerative hierarchical algorithm

- very fast and able to discover clusters of an arbitrary shape (unlike, e.g., *CLARANS*)

The overall process can be described by the following steps:

1. Sample s points from the database D of size n .
2. Divide the sampled s points into p partitions of size s/p each.
3. Cluster each partition independently using the hierarchical merging to k' clusters in each partition. The overall number $k' \cdot p$ of clusters across all partitions is still larger than the user-desired target k .
4. Perform hierarchical clustering over the $k' \cdot p$ clusters derived across all partitions to the user-desired target k .
5. Assign each of the $(n-s)$ non-sample data points to the cluster containing the closest representative.

DATA ANALYSIS

Contents:

- Data standardization techniques
- Range, quartiles, and outliers
- Relationship between the attributes
- Discriminant analysis
- Cluster analysis

Contents:

- Data standardization techniques
- Range, quartiles, and outliers
- Relationship between the attributes
 - Relationship between categorical variables
 - χ^2 - test and Fisher's test
 - Regression analysis (~ relationship between numerical variables)
 - Linear regression for one input variable, correlation coefficients
 - Multi-dimensional linear regression
 - Logistic regression
- Discriminant analysis
- Cluster analysis
 - The main clustering approaches
 - Distance measures
 - k-means clustering
 - Hierarchical clustering
 - Vector quantization
 - Other clustering techniques
 - k-medoids algorithm, *grid-based methods*, and *density-based algorithms*
 - Scalable data clustering approaches